Regular Article

Asymmetric Total Syntheses of Mitragynine, Speciogynine, and 7-Hydroxymitragynine

Jukiya Sakamoto, Mariko Kitajima, and Hayato Ishikawa*

Graduate School of Pharmaceutical Sciences, Chiba University; 1–8–1 Inohana, Chuo-ku, Chiba 260–8675, Japan.
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A number of alkaloids found in *Mitragyna* species belonging to the Rubiaceae family have been shown to have potent biological activity such as analgesic properties. Here, we report the asymmetric total syntheses of mitragynine, speciogynine, and 7-hydroxymitragynine, which are classified as corynantheine-type monoterpenoid indole alkaloids, isolated from *Mitragyna speciosa*. These syntheses were accomplished within 12 steps and in >11% total yield from commercial 3-(trimethylsilyl)propanal using an organocatalytic *anti*-selective Michael reaction and bioinspired transformations.

Key words monoterpenoid indole alkaloid; mitragynine; speciogynine; 7-hydroxymitragynine; bioinspired reaction

Introduction

Mitragyna speciosa, which belongs to the Rubiaceae family, is endemic to the tropics of Southeast Asia, where it is called "Kratom" in Thailand and "Biak Biak" in Malaysia.[1] Traditionally, the leaf of this plant has been used to alleviate the fatigue of workers laboring in the scorching heat and as a substitute for opium. Many chemists have been fascinated by its special biological activity, and isolation and structure elucidation studies of this plant have been conducted since the 1960 s.[2–6] As a result, several corynantheine-type monoterpenoid indole alkaloids have been found with a methoxy group attached to the C-9 position on the indole ring (Fig. 1).

Mitragynine (**1**) is the major component of a base fraction of *Mitragyna speciosa* leaves, and speciogynine (**2**) has been found as the stereoisomer at the C-20 position. Recently, the *in vitro* and *in vivo* activity of **2** at serotonin receptors (5-HTRs) was reported; speciogynine (**2**) exhibited a high affinity for 5-HT1ARs and 5-HT2BRs.[7] 7-Hydroxymitragynine (**3**) was found as a minor component of the leaves; this molecule had a potent analgesic activity *via* opioid μ receptors (mitragynine (**1**) also showed mild analgesic activity).[6,8–16] In addition, it was discovered that this alkaloid maintains potent activity through oral administration.[9] This molecule is now widely recognized as a key component in folk medicine.

Because of their interesting biological activity, several total syntheses of these alkaloids have been reported.[10,17–22] Very recently, our group also achieved the total syntheses of the same class of corynantheine-type monoterpenoid indole alkaloids, corynantheidine and dihydrocorynantheine.[23] However, the efficiency of these total syntheses could be improved, specifically because the goal of previous total syntheses was to reproduce the biosynthetic transformations from strictosidine in the flask.

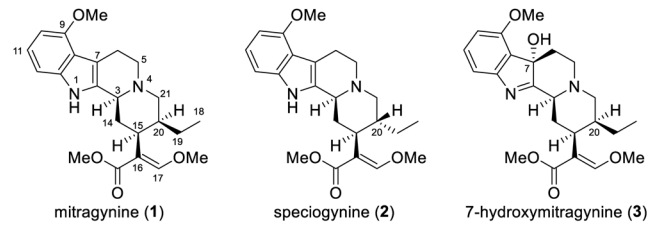


Fig. 1. Structures of Mitragynine, Speciogynine, and 7-Hydroxymitragynine

Here, we report the practical asymmetric total syntheses of mitragynine (**1**), speciogynine (**2**), and 7-hydroxymitragynine (**3**). These syntheses were achieved within 12 steps from commercially available materials with total yields of >11%.

Results and Discussion

Our retrosynthetic analysis of (−)-mitragynine (**1**) and (+)-speciogynine (**2**) is outlined in Chart 1. We envisioned that our previously developed bioinspired transformations toward the syntheses of corynantheine-type alkaloids could also reveal the syntheses of (−)-**1** and its C-20 epimer (+)-**2**. Thus, (−)-**1** and (+)-**2** were retrosynthesized to 9-methoxystrictosidine derivative **5**, *via* bioinspired skeletal transformations and stereoselective reduction of the common intermediate **4**. Preparation of key intermediate **5** was expected to be achieved by diastereoselective Pictet–Spengler cyclization using (*R*)-4-methoxy-α-cyanotryptamine (**6**) and secologanin derivative **7**.[24] Tryptamine derivative **6** was considered for the installation of a 4-methoxyindole unit to the known methyl (*R*)-*N*-benzyloxycarbonyl (Cbz)-azirizine-2-carboxylate (**8**).[25] For the concise synthesis of secologanin derivative **7**, our discovered *anti*-selective organocatalytic Michael reaction was selected as a key step to install all carbons of **7**.[26] Thus, butyraldehyde (**9**) and conjugated ene-yne compound **10**, prepared from commercially available 3-(trimethylsilyl)propanal (**11**), were assumed as optimal building blocks.

As shown in Chart 2, our synthesis commenced with the preparation of (*R*)-4-methoxy-α-cyanotryptamine (**6**). Synthesis of D-tryptophan derivative **13** was performed *via* a Lewis acid-mediated coupling reaction of 4-methoxyindole (**12**) with known chiral aziridine **8**.[27] After several trials, $Yb(OTf)_3$ effectively promoted the coupling reaction and the desired **13** was obtained as a major product (1.5 equivalent (equiv.) of $Yb(OTf)_3$; room temperature (r.t.). Hydrolysis and subsequent

* To whom correspondence should be addressed. e-mail: h_ishikawa@chiba-u.jp

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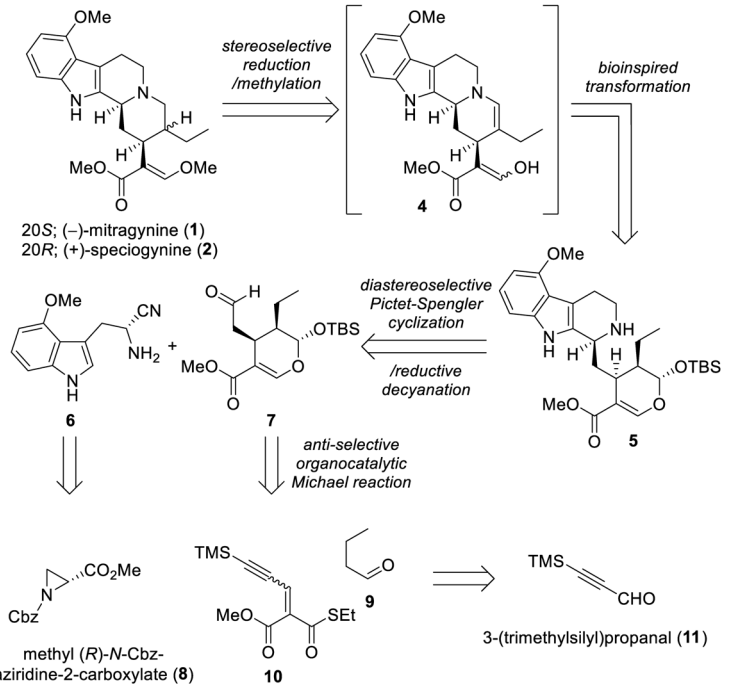


Chart 1. Retrosynthetic Analysis of (−)-Mitragynine and (+)-Speciogynine

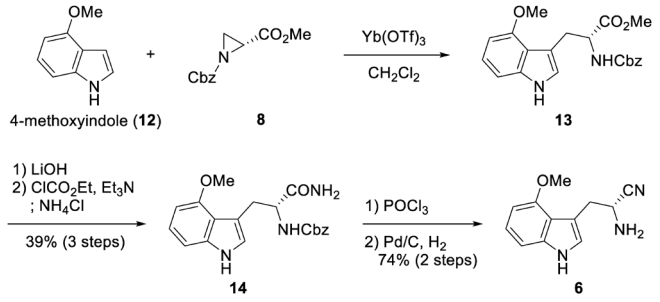


Chart 2. Preparation of (*R*)-4-Methoxy-α-cyanotryptamine

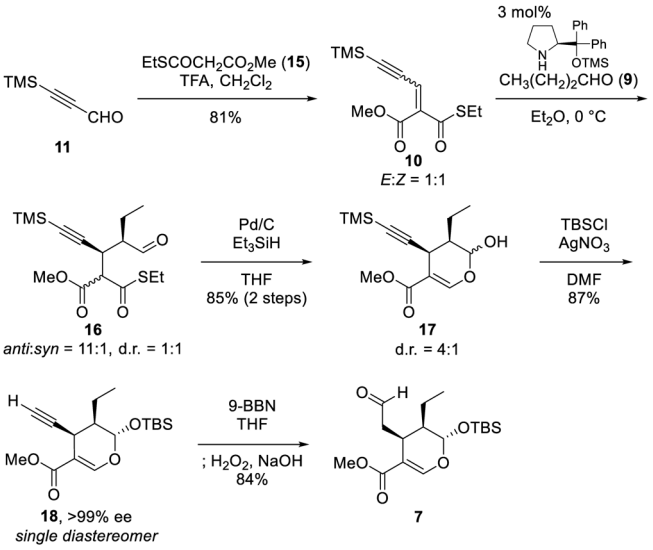


Chart 3. Preparation of Secologanin Derivative

amidation of the crude mixture of **13** gave primary amide **14** in gram quantities (39% yield over three steps). Then, amide **14** was converted to **6** *via* dehydration to construct a cyano group and deprotection of the Cbz group (74% yield over two steps).

As in our previous synthesis of monoterpenoid indole alkaloids, the preparation of secologanin derivative **7** commenced with the Knoevenagel condensation with aldehyde **11** and methyl 3-ethylthio-3-oxopropanoate (**15**)[28] in the presence of trifluoroacetic acid (TFA), affording ene-yne compound **10** (81%, *E*:*Z* = 1 : 1) (Chart 3). To install the C-15 chiral center and C-20 ethyl group in (−)-**1** and (+)-**2**, an *anti*-selective organocatalytic Michael reaction to form **16** was set up. Thus, when ene-yne compound **10** and butyraldehyde (**9**) were treated with 3 mol% diphenylprolinol trimethylsilyl ether catalyst in Et_2O at 0 °C (40 h), the *anti*-adduct **16** was obtained with excellent diastereoselectivity (*anti*:*syn* = 11 : 1). Subsequent Fukuyama reduction[29] to convert aldehyde from a thioester of **16** and spontaneous cyclization were carried out, affording optically active dihydropyran derivative **17** in 85% yield over two steps (4.5 equiv. Et_3SiH, 15 mol% of Pd/C, r.t.). Subsequent silylation of the hemiacetal hydroxy group in **17** was performed. Thus, when **17** was treated with $AgNO_3$ and *tert*-butyldimethylchlorosilane (TBSCl) in dry *N*,*N*-dimethylformamide (DMF), stereoselective silylation proceeded from the sterically less hindered site to afford α-oriented silyl ether **18** as a single isomer (87%). Enantiomeric excess was determined at this stage: >99% enantiomeric excess (ee) by chiral HPLC analysis. Finally, alkyne **18** was treated with hydroboration–oxidation conditions to afford secologanin derivative **7** in excellent yield (84%, 1.2 equiv. 9-borabicyclo[3.3.1]nonane (9-BBN), tetrahydrofuran (THF), r.t. then 5.0 equiv. H_2O_2, 1.0 equiv. NaOH, r.t.). All protocols in the synthesis of **7** were performed in gram-scale quantities.

With the desired synthons **6** and **7** in hand, we next attempted bioinspired transformations for the construction of a corynantheine-type scaffold (Chart 4). To prepare 9-methoxystrictosidine derivative **5**, our developed diastereoselective

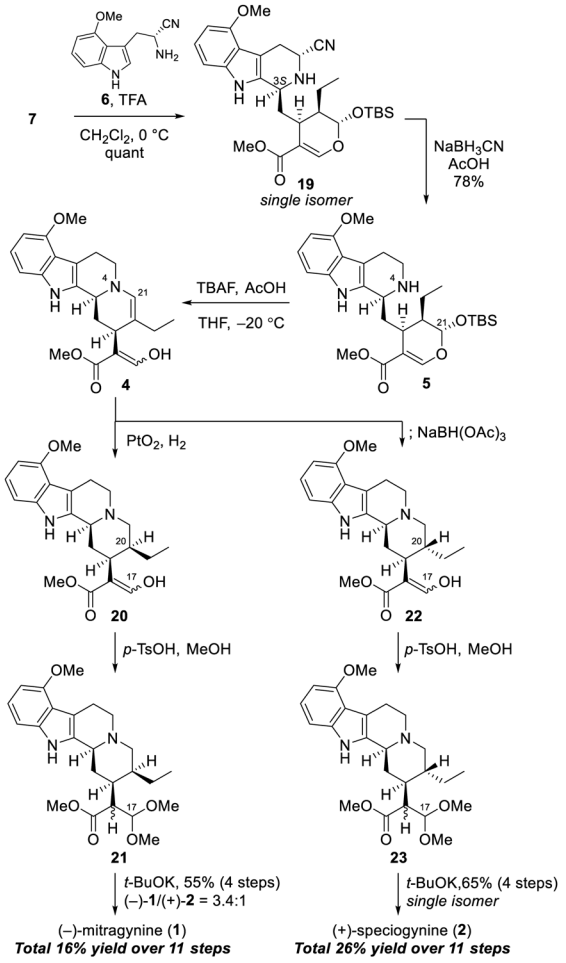


Chart 4. Total Syntheses of (−)-Mitragynine and (+)-Speciogynine

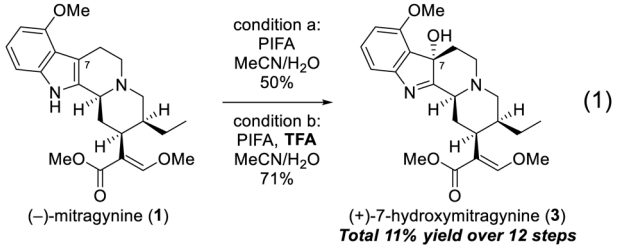
Pictet–Spengler cyclization/reductive decyanation sequences were examined. When a mixture of **6** and **7** was treated with 50 mol% TFA in CH_2Cl_2 at 0 °C (30 min), the desired 3*S*-isomer **19** was obtained as a single diastereomer in quantitative yield.[30] Reductive removal of the cyano group in **19** was performed by treatment with $NaBH_3CN$ in the presence of AcOH in MeOH, affording 9-methoxystrictosidine derivative **5** in excellent yield (78%).

Next, key bioinspired transformations triggered by removal of the silyl group in **5** were examined. Thus, **5** was exposed to a simple desilylation condition (2.0 equiv. tetrabutylammonium fluoride (TBAF), 2.0 equiv. AcOH, THF, –20 °C) to afford enamine **4** linked by N-4 and C-21. After confirming a full conversion to **4** from **5** by NMR spectroscopic analysis, the crude mixture of **4** was directly treated with 100 wt% PtO_2 under H_2 atmosphere.[17] As a result, kinetically reduced product **20**, which has the same C-20*S* stereochemistry as (−)-mitragynine (**1**), was preferentially obtained. To complete the total synthesis of (−)-**1**, modification of the enol hydroxy group at the C-17 position was performed. Thus, the crude mixture of **20** was treated with *p*-TsOH in MeOH to afford dimethoxy acetal **21**.[31] Finally, stereoselective E1cB elimination of the C-17 methoxy group of **21**, using *t*-BuOK in DMF,[17] afforded a separable mixture of (−)-mitragynine (**1**) and (+)-speciogynine (**2**) [(−)-**1**/(+)-**2** = 3.4 : 1, 55% yield over four steps].

Bioinspired transformations toward the selective synthesis of C-20*R* isomer (+)-speciogynine (**2**) were also demonstrated. After generating enamine **4** under the same conditions as for the synthesis of (−)-**1**, 2.0 equiv. $NaBH(OAc)_3$ was added in a one-pot protocol. In this case, thermodynamic controlled protonation at the C-20 position followed by reduction of a generated iminium ion afforded C-20*R* isomer **22** as a single diastereomer. Subsequent modification of the C-17 enol hydroxy group was carried out as in the synthesis of (−)-**1**, and the E1cB elimination reaction, following the formation of **23**, afforded (+)-speciogynine (**2**) in 65% overall yield within four steps from **5**.

In summary, practical total syntheses of (−)-mitragynine (**1**, total 16% over 11 steps from **11**) and (+)-speciogynine (**2**, total 26% over 11 steps from **11**) were accomplished.

Next, to prepare (+)-7-hydroxymitragynine (**3**), the direct C-7 oxidation of (−)-**1** was investigated (Eq. 1). Following our previously reported method, when freshly purified (−)-**1** was oxidized in aqueous acetonitrile by treatment with 1.0 equiv. [bis(trifluroacetoxy)iodo]benzene (PIFA), (+)-**3** was obtained in moderate yield (condition a in Eq. 1, 50%).[32] After several trials toward further improvement of the chemical yield, it was determined that the addition of 2.0 equiv. TFA led to the effective production of (+)-**3** (condition b in Eq. 1, 71%). We suppose that TFA masked the reactive lone pair of the N-4 nitrogen in (−)-**1** as a salt, thereby suppressing undesired N-4 oxidation. As a result, the total synthesis of (+)-7-hydroxymitragynine (**3**) was accomplished with 11% overall yield within 12 steps from **11.**



(1)

Conclusion

In conclusion, we accomplished the total syntheses of three bioactive *Mitragyna* alkaloids: (−)-mitragynine (**1**), (+)-speciogynine (**2**), and (+)-7-hydroxymitragynine (**3**). The synthesis of secologanin derivative **7**, a key fragment in this synthesis, was carried out *via* an *anti*-selective organocatalytic Michael reaction using ene-yne compound **10** and simple aldehyde **9**, with excellent enantiomeric excess. Preparation of common intermediate **4** was achieved *via* bioinspired transformations of strictosidine derivative **5** triggered by the removal of a silyl group, which branched into (−)-**1** and (+)-**2** by stereoselective reduction. In addition, (+)-**3** was obtained through direct oxidation of (−)-**1** *via* a nitrogen-masking strategy. As a result, syntheses of (−)-mitragynine (**1**), (+)-speciogynine (**2**), and (+)-7-hydroxymitragynine (**3**) were realized within fewer than 12 steps and in >11% overall yield. Using the synthesized natural products and their derivatives, a study of their structure–activity relationships is currently under way in our laboratory.

Experimental

General Experimental Procedures All reactions were monitored by TLC using Merck 60 F254 precoated silica gel

plates (0.25 mm thick) and Fuji Silysia Chemical precoated amino-silica gel plates (0.25 mm thick). UV spectra were recorded in MeOH on a JASCO V-560 instrument. Specific optical rotations were measured using a JASCO P-2200 polarimeter. Circular dichroism spectra were recorded on a JASCO J-1100 spectrometer. Fourier transform (FT) IR spectra were recorded on a JASCO FT/IR-4700. ^{1}H- and ^{13}C-NMR spectra were recorded on a JEOL ECZ 400 (400 MHz ^{1}H-NMR, 100 MHz for ^{13}C-NMR), ECS 400 (400 MHz ^{1}H-NMR, 100 MHz for ^{13}C-NMR), JEOL ECX 500 (500 MHz ^{1}H-NMR, 125 MHz for ^{13}C-NMR), and ECZ 600 (600 MHz for ^{1}H-NMR, 150 MHz for ^{13}C-NMR) FT-NMR spectrometer instrument. Data for ^{1}H-NMR are reported as chemical shifts (δ ppm), multiplicity (s = singlet, d = doublet, t = triplet, dd = double doublet, ddd = double double doublet, dt = double triplet, q = quartet, m = multiplet, br = broad), coupling constant (Hz), integration, and assignment. Data for ^{13}C-NMR are reported as chemical shifts. The high-resolution mass spectra were recorded on a JEOL AccuTOF LC-plus JMS-T100LP. Flash chromatography was performed using Kanto Chemical silica gel 60N and Fuji Silysia Chemical amino-silica gel (SiO_2-NH, NH-DM2035). HPLC analysis was performed on a JASCO MD-4017; UV detection was monitored at an appropriate wavelength using Daicel Chiralcel OD-H (0.46 × 25 cm).

Preparation of Compound 14 To a solution of known aziridine **8** (1.0 g, 4.25 mmol) and 4-methoxyindole (**12**, 1.25 g, 8.50 mmol) in CH_2Cl_2 (17 mL) was added $Yb(OTf)_3$ (2.64 g, 4.25 mmol) under Ar atmosphere. The reaction mixture was stirred for 24 h at r.t. under Ar atmosphere in the dark. To the resulting mixture was added additional $Yb(OTf)_3$ (1.32 g, 2.13 mmol), followed by stirring for a further 48 h at r.t. The mixture was then quenched with saturated aqueous $NaHCO_3$ and the resulting mixture was filtered through a Celite pad, eluted with $CHCl_3$. The filtrate was extracted three times with $CHCl_3$. The combined organic layer was dried over Na_2SO_4 and then concentrated under reduced pressure. Flash chromatography (SiO_2, 22–28% AcOEt/*n*-hexane gradient) provided an inseparable regioisomer mixture of **13** (780 mg, r.r. = 6.7 : 1). To a solution of a mixture of **13** in aqueous THF (34 mL, THF/H_2O = 10 : 1) was added 2 M aqueous LiOH (3.06 mL, 6.12 mmol) at r.t. under Ar atmosphere. The reaction mixture was stirred for 2.5 h at r.t. before quenching with 2 M aqueous HCl (pH 2). Brine was added to the resulting mixture and the aqueous layer was extracted three times with AcOEt. The combined organic layer was dried over Na_2SO_4, and concentrated under reduced pressure. Obtained crude materials were dissolved in anhydrous THF (40.8 mL), then ethyl chloroformate (272 μL, 2.86 mmol) and triethylamine (853 μL, 6.12 mmol) were added at 0 °C under Ar atmosphere. The reaction mixture was stirred for 30 min at 0 °C. To the resulting mixture was added 1.0 M aqueous NH_4Cl (3.06 mL, 3.06 mmol) at 0 °C. The reaction mixture was stirred for an additional 20 min at 0 °C. It was then quenched with saturated aqueous $NaHCO_3$ and the aqueous layer was extracted three times with AcOEt. The combined organic layer was dried over Na_2SO_4, and concentrated under reduced pressure. Flash chromatography (SiO_2, 2%–5% MeOH/$CHCl_3$ gradient) afforded amide **14** (607.5 mg, 39% over three steps).

Compound **14**: White amorphous powder; $[\alpha]_D^{23}$ +2.6 (*c* 0.31, MeOH); IR (ATR) ν_{max} cm^{-1} 3448, 3338, 2945, 2480, 1674, 1620, 1589, 1525, 1502, 1423, 1356, 1313, 1257, 1153, 1095, 1039, 966, 910, 860, 775, 752, 735, 719, 694, 617; high resolution (HR)MS (electrospray ionization (ESI)) $[M + Na]^+$ Calcd. for $[C_{20}H_{21}N_3Na_1O_4]^+$: 390.1430, Found: 390.1435; UV (MeOH) λ_{max} 291, 281, 265, 220 nm; ^{1}H-NMR (400 MHz, $(CD_3)_2SO$) δ: 10.8 (br s, 1H), 7.34–7.19 (m, 5H), 7.03 (br s, 1H), 6.96–6.92 (m, 3H), 6.44 (d, J = 6.4 Hz, 1H), 4.95 (s, 2H), 4.23 (dt, J = 8.8, 4.4 Hz, 1H), 3.84 (s, 3H), 3.28 (dt, J = 14.4, 3.6 Hz, 1H), 2.98 (ddd, J = 14.4, 10.0, 3.2 Hz, 1H); ^{13}C-NMR (150 MHz, $(CD_3)_2SO$) δ: 174.2, 155.8, 154.1, 137.8, 137.1, 128.4, 127.7, 127.4, 122.2, 121.8, 117.0, 110.6, 105.0, 98.9, 65.2, 56.2, 55.1, 39.9, 39.8, 39.7, 39.5, 39.4, 39.2, 39.1, 29.3.

Preparation of (*R*)-4-Methoxy-α-cyanotryptamine (6) To a solution of amide **14** (608 mg, 1.65 mmol) in anhydrous pyridine (11 mL) was added phosphoryl chloride (355 μL, 3.80 mmol) at 0 °C under Ar atmosphere. The reaction mixture was stirred for 50 min at 0 °C. MeOH was added at 0 °C to the resulting mixture and stirred for 10 min, followed by concentration under reduced pressure. The resulting mixture was dissolved in Et_2O, and the organic layer was washed with 1 M aqueous NaOH three times, and with brine, dried over Na_2SO_4, and then concentrated under reduced pressure. Flash chromatography (SiO_2, 1% MeOH/$CHCl_3$) gave a dehydrated compound (510 mg, 88%). To a solution of the dehydrated compound (500 mg, 1.43 mmol) in degassed 1,4-dioxane (10.4 mL) was added 10% Pd/C (152 mg, 0.143 mmol) at r.t. under Ar atmosphere. The resulting mixture was purged with a stream of hydrogen and the reaction mixture was stirred for 13 h at r.t. under H_2 atmosphere. The resulting mixture was filtered with a Celite pad with $CHCl_3$. The filtrate was concentrated under reduced pressure and the resulting residue was purified by flash chromatography (SiO_2, 60% AcOEt/*n*-hexane, then 5% MeOH/$CHCl_3$) to afford (*R*)-4-methoxy-α-cyanotryptamine (**6**, 260.0 mg, 84%).

(*R*)-4-Methoxy-α-cyanotryptamine (**6**): Pale pink amorphous powder; $[\alpha]_D^{23}$ –15.0 (*c* 1.96, $CHCl_3$); IR (ATR) ν_{max} cm^{-1} 3363, 3194, 2914, 1741, 1587, 1547, 1510, 1431, 1356, 1257, 1167, 1132, 1086, 1053, 955, 895, 852, 814, 683, 623; HRMS (ESI) $[M–CN]^+$ Calcd. for $[C_{11}H_{13}N_2O_1]^+$: 189.1028, Found: 189.1025; UV (MeOH) λ_{max} 290, 281, 267, 220 nm; ^{1}H-NMR (600 MHz, $CDCl_3$) δ: 8.22 (br s, 1H), 7.11 (t, J = 7.8 Hz, 1H), 7.00 (d, J = 2.4 Hz, 1H), 6.98 (d, J = 7.8 Hz, 1H), 6.52 (d, J = 7.8 Hz, 1H), 4.18 (t, J = 7.2 Hz, 1H), 3.93 (s, 3H), 3.33 (dd, J = 13.8, 7.2 Hz, 1H), 3.27 (dd, J = 13.8, 7.2 Hz, 1H); ^{13}C-NMR (150 MHz, $CDCl_3$) δ: 154.4, 138.2, 123.3, 122.6, 122.5, 117.0, 110.0, 104.9, 99.8, 55.3, 45.4, 33.6.

Preparation of Compound 17 To the solution of an *E* and *Z* mixture of half thioester malonate derivative **10** (500 mg, 1.85 mmol) and diphenylprolinol trimethyl silyl ether (18.0 mg, 0.0555 mmol) in Et_2O (9.2 mL) was added *n*-butyraldehyde (**9**, 167 μL, 1.85 mmol) at 0 °C under Ar atmosphere. The reaction mixture was stirred for 40 h at 0 °C. The resulting mixture was concentrated under reduced pressure. Crude ^{1}H-NMR revealed that *anti*-adduct **16** was the main product (diastereomer ratio of **16**; *anti:syn* = 11 : 1). The crude materials of **16** were solved to degassed THF (3.7 mL), then 10% Pd/C (197 mg, 0.185 mmol) and Et_3SiH (886 μL, 5.55 mmol) were added at 0 °C under Ar atmosphere. The reaction mixture was stirred for 4.5 h at r.t, whereafter an additional 10% Pd/C (98.4 mg, 0.0924 mmol) and Et_3SiH (443 μL, 2.77 mmol) were added at r.t. under Ar atmosphere. This was stirred for 5 h at r.t. before the addition of 1 M aqueous HCl (3.7 mL). The resulting

mixture was stirred for a further 18 h at r.t. before filtration with a Celite pad eluted with AcOEt, and partitioned between an AcOEt layer and an aqueous layer. The aqueous layer was extracted three times with AcOEt. The combined organic layer was dried over $MgSO_4$, and concentrated under reduced pressure. Flash chromatography (SiO_2, 5%–20% AcOEt/*n*-hexane gradient) afforded two diastereomers of dihydropyran **17** (444.5 mg, 85% over two steps, d.r. = 4 : 1).

Compound **17**: Yellow waxy solid; IR (ATR) ν_{max} cm^{-1} 3370, 2953, 1672, 1629, 1444, 1383, 1311, 1247, 1204, 1148, 1125, 1100, 1030, 1015, 981, 945, 908, 879, 835; HRMS (ESI) $[M+Na]^+$ Calcd. for $[C_{14}H_{22}Na_1O_4Si_1]^+$: 305.1185, Found: 305.1145; ^{1}H-NMR (500 MHz, $CDCl_3$) δ: 7.53 (s, 1H), 7.46 (s, 1H), 5.36 (dd, J = 12.0, 1.5 Hz, 1H), 5.22 (t, J = 7.5 Hz, 1H), 4.85 (d, J = 7.5 Hz, 1H), 3.77 (s, 3H), 3.75 (s, 3H), 3.58 (d, J = 5.0 Hz, 1H), 3.55 (d, J = 5.0 Hz, 1H), 3.34 (d, J = 7.5 Hz, 1H), 1.85–1.71 (m, 3H), 1.64–1.53 (m, 3H), 1.05 (t, J = 7.0 Hz, 3H), 1.02 (t, J = 7.0 Hz, 3H), 0.14 (s, 9H), 0.12 (s, 9H); ^{13}C-NMR (125 MHz, $CDCl_3$) δ: 166.9, 166.7, 153.3, 152.5, 107.2, 106.3, 104.0, 103.7, 97.1, 96.3, 90.1, 97.9, 51.7, 51.6, 42.2, 39.9, 26.8, 23.6, 21.6, 20.4, 11.4, 11.0, 0.17 (3C), 0.12 (3C).

Preparation of Compound 18 To a solution of two diastereomers of dihydropyran **17** (1.0 g, 3.54 mmol) in dry DMF (11.8 mL) were added $AgNO_3$ (782 mg, 4.60 mmol) and TBSCl (800.5 mg, 5.31 mmol) at 0 °C under Ar atmosphere. The reaction mixture was stirred for 1 h at r.t. under Ar atmosphere. Additional $AgNO_3$ (391 mg, 2.30 mmol) and TBSCl (400 mg, 2.66 mmol) were then added under Ar atmosphere. The reaction mixture was stirred for 2.5 h at r.t. Additional $AgNO_3$ (391 mg, 2.30 mmol) was then added under Ar atmosphere and the mixture was stirred for 15 min at r.t. The resulting mixture was directly filtered through a short plug of silica gel eluted with 20% AcOEt/*n*-hexane and the filtrate was added to saturated $NaHCO_3$. The aqueous layer was extracted three times with AcOEt. The combined organic layer was washed four times each with water and brine, dried over $MgSO_4$, and concentrated under reduced pressure. Flash chromatography (SiO_2, 1% AcOEt/*n*-hexane) afforded **18** (1.0 g, 87%). Enantiomeric excess was determined to be over 99% by HPLC (Chiralcel OD-H column) (see Supporting Information). Results for compound **18**: 1.5% *i*-PrOH/*n*-hexane, 0.500 mL/min; major enantiomer t_R = 7.27 min, minor enantiomer t_R = 8.30 min.

Compound **18**: Colorless oil; $[\alpha]^{22}_D$ –235.6 (*c* 0.71, $CHCl_3$); IR (ATR) ν_{max} cm^{-1} 3315, 3269, 2954, 2931, 2883, 2858, 1714, 1639, 1462, 1437, 1392, 1298, 1254, 1228, 1167, 1130, 1088, 1001, 949, 924, 879, 837, 781, 764, 679, 663, 638, 617; HRMS (ESI) $[M+Na]^+$ Calcd. for $[C_{17}H_{28}Na_1O_4Si_1]^+$: 347.1655, Found: 347.1669; ^{1}H-NMR (400 MHz, $CDCl_3$) δ: 7.48 (s, 1H), 5.18 (d, J = 8.4 Hz, 1H), 3.75 (s, 3H), 3.56 (dd, J = 5.2, 2.4 Hz, 1H), 2.15 (d, J = 2.4 Hz, 1H), 1.78 (m, 1H), 1.59 (m, 1H), 1.47 (m, 1H), 1.00 (t, J = 7.2 Hz, 3H), 0.93 (s, 9H), 0.15 (s, 3H), 0.15 (s, 3H); ^{13}C-NMR (100 MHz, $CDCl_3$) δ: 167.1, 153.9, 106.6, 97.6, 82.4, 71.4, 51.6, 42.5, 25.8, 20.3, 18.1, 11.1, –4.1, –5.1.

Preparation of Secologanin Derivative 7 To a solution of alkyne **18** (1.0 g, 3.1 mmol) in anhydrous THF (10.3 mL) was added 9-BBN (1.0 M in THF solution, 7.4 mL, 3.70 mmol) under Ar atmosphere. The reaction mixture was stirred for 2.5 h at r.t. The resulting mixture was quenched with H_2O (1.0 mL) and then stirred for an additional 10 min. To the resulting mixture were added 30% aqueous H_2O_2 (1.57 mL, 15.4 mmol) and 1.0 M aqueous NaOH (3.08 mL, 3.08 mmol) at 0 °C. This reaction mixture was stirred for 80 min at r.t. under Ar atmosphere. After the addition of saturated aqueous NH_4Cl, the aqueous layer was extracted three times with AcOEt. The combined organic layer was washed with brine, dried over $MgSO_4$, and concentrated under reduced pressure. Flash chromatography, repeated three times (SiO_2, 5%–50% AcOEt/*n*-hexane gradient then 10% MeOH/$CHCl_3$) afforded secologanin derivative **7** (887 mg, 84%).

Secologanin Derivative **7**: Colorless oil; $[\alpha]^{23}_D$ –141.1 (*c* 1.89, $CHCl_3$); IR (ATR) ν_{max} cm^{-1} 2954, 2931, 2887, 2858, 1703, 1633, 1466, 1439, 1385, 1362, 1309, 1281, 1254, 1167, 1138, 1101, 1080, 1007, 918, 874, 837, 781, 675, 640; HRMS (ESI) $[M+Na]^+$ Calcd. for $[C_{17}H_{30}Na_1O_5Si_1]^+$: 365.1760, Found: 365.1769; ^{1}H-NMR (400 MHz, $CDCl_3$) δ: 9.76 (dd, J = 3.2, 2.0 Hz, 1H), 7.49 (s, 1H), 5.06 (d, J = 7.6 Hz, 1H), 3.69 (s, 3H), 3.30 (dt, J = 7.6, 5.2 Hz, 1H), 2.48 (ddd, J = 15.2, 5.2, 2.0 Hz, 1H), 2.35 (ddd, J = 15.2, 7.6, 3.2 Hz, 1H), 1.74–1.59 (m, 2H), 1.15 (m, 1H), 0.98 (t, J = 7.2 Hz, 3H), 0.91 (s, 9H), 0.14 (s, 3H), 0.13 (s, 3H); ^{13}C-NMR (150 MHz, $CDCl_3$) δ: 201.6, 167.7, 154.0, 108.5, 96.4, 51.4, 45.1, 43.6, 26.8, 25.8, 19.3, 18.1, 11.5, –4.2, –5.1.

Preparation of Compound 19 To a solution of secologanin derivative **7** (320 mg, 0.934 mmol), (*R*)-4-methoxy-α-cyanotryptamine (**6**, 201.1 mg, 0.934 mmol), and powdered MS 4 Å (1 g) in CH_2Cl_2 (18.7 mL) was added TFA (1.0 M in CH_2Cl_2, 467 μL, 0.467 mmol) at 0 °C under Ar atmosphere. The reaction mixture was stirred for 30 min at 0 °C. The resulting mixture was filtered through a cotton plug with $CHCl_3$ and the filtrate was added to an excess amount of saturated aqueous $NaHCO_3$ solution at 0 °C. The aqueous layer was extracted three times with $CHCl_3$. The combined organic layer was dried over Na_2SO_4, and concentrated under reduced pressure. Flash chromatography (SiO_2, 15% AcOEt/*n*-hexane) afforded **19** (504.3 mg, quantitative yield) as a single diastereomer.

Compound **19**: White amorphous powder; $[\alpha]^{23}_D$ –230.1 (*c* 0.85, $CHCl_3$); IR (ATR) ν_{max} cm^{-1} 3328, 2928, 1688, 1630, 1572, 1509, 1463, 1437, 1384, 1357, 1312, 1279, 1254, 1207, 1167, 1137, 1102, 1029, 1018, 974, 938, 892, 871, 837; HRMS (ESI) $[M+H]^+$ Calcd. for $[C_{29}H_{42}N_3O_5Si_1]^+$: 540.2894, Found: 540.2859; UV (MeOH) λ_{max} 292, 225 nm; ^{1}H-NMR (600 MHz, $CDCl_3$) δ: 7.82 (br s, 1H), 7.59 (s, 1H), 7.04 (t, J = 7.8 Hz, 1H), 6.91 (d, J = 7.8 Hz, 1H), 6.47 (d, J = 7.8 Hz, 1H), 5.26 (d, J = 9.0 Hz, 1H), 4.37 (dd, J = 4.2, 3.6 Hz, 1H), 4.07 (ddd, J = 10.8, 3.6, 1.8 Hz, 1H), 3.88 (s, 3H), 3.79 (s, 3H), 3.35–3.29 (m, 2H), 2.97 (dt, J = 12.0, 4.2 Hz, 1H), 1.79 (ddd, J = 12.6, 11.4, 3.0 Hz, 1H), 1.73–1.63 (m, 2H), 1.55 (td, J = 12.6, 2.4 Hz, 1H), 1.18 (m, 1H), 0.98 (t, J = 7.2 Hz, 3H), 0.93 (s, 9H), 0.19 (s, 3H), 0.18 (s, 3H); ^{13}C-NMR (150 MHz, $CDCl_3$) δ: 169.3, 154.6, 154.4, 137.4, 132.4, 122.8, 120.0, 117.5, 109.0, 105.6, 104.3, 100.0, 97.2, 55.3, 52.2, 46.8, 44.4, 43.8, 36.9, 27.8, 26.9, 25.8, 20.2, 18.1, 11.1, –3.9, –4.9.

Preparation of 9-Methoxystrictosidine Derivative 5 Acetic acid (265 μL, 4.63 mmol) and sodium cyanoborohydride (582 mg, 9.26 mmol) were added to a solution of **19** (500 mg, 0.936 mmol) in MeOH (4.6 mL) at r.t. under Ar atmosphere. The reaction mixture was stirred for 24 h at r.t. To the resulting mixture were added additional acetic acid (265 μL, 4.63 mmol) and sodium cyanoborohydride (582 mg, 9.26 mmol) under Ar atmosphere. The reaction mixture was stirred for a further 36 h at r.t. The resulting mixture was quenched with

saturated aqueous $NaHCO_3$ solution at 0 °C. The aqueous layer was extracted three times with AcOEt. The combined organic layer was washed with brine, dried over Na_2SO_4, and concentrated under reduced pressure. The crude materials were filtered through a short plug of amino-silica gel (SiO_2-NH), eluted with 5% MeOH/$CHCl_3$, to remove an excess amount of sodium cyanoborohydride. Flash chromatography (SiO_2, 25% AcOEt/*n*-hexane then 20% MeOH/$CHCl_3$) afforded the 9-methoxystrictosidine derivative **5** (371.6 mg, 78%) with 80.5 mg of starting material **19**.

9-Methoxystrictosidine Derivative **5**; White amorphous powder; $[\alpha]_D^{23}$ –189.9 (*c* 0.93, $CHCl_3$); IR (ATR) ν_{max} cm^{-1} 2933, 2850, 1698, 1630, 1510, 1460, 1436, 1383, 1357, 1308, 1276, 1252, 1166, 1137, 1096, 1031, 1007, 975, 950, 927, 908, 872, 861, 836, 825; HRMS (ESI) $[M+H]^+$ Calcd. for $[C_{28}H_{43}N_2O_5Si_1]^+$: 515.2941, Found: 515.2955; UV (MeOH) λ_{max} 292, 223 nm; ^{1}H-NMR (600 MHz, $CDCl_3$) δ: 8.33 (br s, 1H), 7.55 (s, 1H), 7.02 (t, J = 7.8 Hz, 1H), 6.94 (d, J = 7.8 Hz, 1H), 6.46 (d, J = 7.8 Hz, 1H), 5.23 (d, J = 8.4 Hz, 1H), 4.01 (dd, J = 9.0, 3.0 Hz, 1H), 3.88 (s, 3H), 3.74 (s, 3H), 3.29 (m, 1H), 3.05–2.95 (m, 4H), 1.92 (ddd, J = 13.8, 10.8, 3.6 Hz, 1H), 1,71–1.61 (m, 2H), 1.52 (ddd, J = 13.8, 10.2, 3.6 Hz, 1H), 1.27 (m, 1H), 0.99 (t, J = 7.2 Hz, 3H), 0.92 (s, 9H), 0.17 (s, 3H), 0.16 (s, 3H); ^{13}C-NMR (150 MHz, $CDCl_3$) δ: 169.3, 154.8, 154.6, 137.3, 133.6, 122.3, 117.5, 109.8, 108.6, 104.5, 99.9, 97.3, 55.4, 51.8, 50.5, 44.4, 42.1, 36.7, 28.0, 25.8, 24.2, 20.1, 18.1, 11.6, –4.0, –5.0.

Synthesis of (−)-Mitragynine (1) To a solution of 9-methoxystrictosidine derivative **5** (150 mg, 0.291 mmol) in degassed anhydrous THF (2.9 mL) were added acetic acid (83.3 μL, 1.46 mmol) and tetrabutylammonium fluoride (1.0 M in THF solution, 583 μL, 0.583 mmol) at –20 °C under Ar atmosphere. The reaction mixture was stirred for 5 h at –20 °C. The resulting mixture was directly filtered through a short plug of silica gel eluted with 40% AcOEt/*n*-hexane. The residue of **4** was dissolved in degassed MeOH (14.6 mL) to which was added PtO_2 (150 mg, 100 wt%) at r.t. under Ar atmosphere. The resulting mixture was purged with a stream of hydrogen and the reaction mixture was stirred for 1 h at r.t. under H_2 atmosphere. The resulting mixture was filtered with a Celite pad with AcOEt, and the filtrate was concentrated under reduced pressure. The resulting residue of **20** was dissolved in degassed MeOH (14.6 mL) to which was added $CH(OMe)_3$ (52.7 μL, 0.481 mmol) and *p*-TsOH·H_2O (211 mg, 1.11 mmol), and the mixture was stirred at 70 °C for 16 h. The resulting mixture was quenched with saturated aqueous $NaHCO_3$ solution, and was concentrated under reduced pressure. The aqueous layer was extracted three times with $CHCl_3$. The combined organic layer was dried over Na_2SO_4, and concentrated under reduced pressure. The crude materials of **21** were dissolved in dry degassed DMF (5.8 mL) to which was added *t*-BuOK (98.1 mg, 0.874 mmol) at 0 °C under Ar atmosphere. The reaction mixture was stirred for 45 min at r.t. under Ar atmosphere. The resulting mixture was quenched with water and the aqueous layer was extracted three times with $CHCl_3$. The combined organic layer was dried over Na_2SO_4, and concentrated under reduced pressure. Flash chromatography (SiO_2, 25–45% AcOEt/*n*-hexane gradient then 5% MeOH/$CHCl_3$) afforded (−)-mitragynine (**1**, 48.7 mg, 42% over four steps) and (+)-speciogynine (**2**, 15.2 mg, 13% over four steps). All spectral data of the obtained (−)-mitragynine (**1**) and (+)-speciogynine (**2**) were identical to those of natural products.[33,34]

(−)-Mitragynine (**1**): Yellow amorphous powder; $[\alpha]_D^{24}$ –103.1 (*c* 0.57, $CHCl_3$) [lit. $[\alpha]_D^{24}$ –126 (*c* 1.2, $CHCl_3$)]; IR (ATR) ν_{max} cm^{-1} 3359, 2926, 1697, 1622, 1567, 1508, 1461, 1434, 1350, 1273, 1253, 1188, 1170, 1146, 1104, 1078, 1020, 977, 903, 885, 862; HRMS (ESI) $[M+H]^+$ calculated for $[C_{23}H_{31}N_2O_4]^+$: 399.2284, found: 399.2291; CD (0.3 mM, MeOH, 23 °C) λ nm ($\Delta\varepsilon$): 320 (0), 291 (2.27), 283 (2.15), 273 (2.45), 261 (0), 239 (–8.18), 229 (–7.33), 219 (–11.32), 206 (0); UV (MeOH) λ_{max} 292, 224 nm; ^{1}H- and ^{13}C-NMR see supporting information.

(+)-Speciogynine (**2**): Yellow amorphous powder; $[\alpha]_D^{24}$ +17.0 (*c* 1.02, $CHCl_3$) [lit. $[\alpha]_D^{24}$ +26.8 (*c* 0.85, $CHCl_3$)]; IR (ATR) ν_{max} cm^{-1} 3363, 2931, 2846, 2796, 2736, 1699, 1633, 1568, 1508, 1460, 1434, 1355, 1335, 1317, 1276, 1252, 1222, 1187, 1144, 1102, 1013, 989, 915, 894, 862, 834, 810; HRMS (ESI) $[M+H]^+$ Calcd. for $[C_{23}H_{31}N_2O_4]^+$: 399.2284, Found: 399.2273; CD (0.3 mM, MeOH, 23 °C) λ nm ($\Delta\varepsilon$): 309 (0), 291 (2.93), 281 (3.56), 269 (4.47), 253 (3.36), 237 (9.55), 230 (0), 219 (–20.57), 203 (0); UV (MeOH) λ_{max} 291, 225 nm; ^{1}H- and ^{13}C-NMR see supporting information.

Synthesis of (+)-Speciogynine (2) To a solution of 9-methoxystrictosidine derivative **5** (30 mg, 0.058 mmol) in degassed anhydrous THF (580 μL) were added acetic acid (16.7 μL, 0.291 mmol) and tetrabutylammonium fluoride (1.0 M in THF solution, 117 μL, 0.117 mmol) at –20 °C under Ar atmosphere. The reaction mixture was stirred for 5 h at –20 °C. To the resulting mixture was added sodium triacetoxyborohydride (24.7 mg, 0.117 mmol) at –20 °C. The reaction mixture was stirred for 2.5 h at –20 °C. The resulting mixture was quenched with saturated aqueous $NaHCO_3$ solution and the aqueous layer was extracted three times with AcOEt. The combined organic layer was washed with brine then dried over Na_2SO_4. The resulting residue of **22** was dissolved in degassed MeOH (2.9 mL), to which was added $CH(OMe)_3$ (10.5 μL, 0.0962 mmol) and *p*-TsOH·H_2O (42.1 mg, 0.221 mmol), and then stirred at 70 °C for 12 h. The resulting mixture was quenched with saturated aqueous $NaHCO_3$ solution and then concentrated under reduced pressure. The aqueous layer was extracted three times with $CHCl_3$. The combined organic layer was dried over Na_2SO_4, and concentrated under reduced pressure. The crude materials of **23** were dissolved in dry degassed DMF (1.2 mL) to which was added *t*-BuOK (19.6 mg, 0.175 mmol) at 0 °C under Ar atmosphere. The reaction mixture was stirred for 40 min at r.t. under Ar atmosphere. The resulting mixture was quenched with water and the aqueous layer was extracted five times with $CHCl_3$. The combined organic layer was dried over Na_2SO_4, and concentrated under reduced pressure. Flash chromatography (SiO_2, 45% AcOEt/*n*-hexane) afforded (+)-speciogynine (**2**, 15.2 mg, 65% over four steps).

Synthesis of (+)-7-Hydroxymitragynine (3) To a solution of freshly purified (−)-mitragynine (**1**, 30 mg, 0.075 mmol) in MeCN (900 μL) and water (100 μL) was added TFA (11.5 μL, 0.151 mmol) at 0 °C, and the reaction mixture was stirred for 5 min at 0 °C. PIFA (32.4 mg, 0.0753 mmol) was added to the stirred mixture at 0 °C under Ar atmosphere. The reaction mixture was stirred for 20 min at 0 °C, followed by quenching with saturated aqueous $NaHCO_3$ solution. The aqueous layer was extracted three times with CH_2Cl_2. The combined organic layer was washed with brine, dried over Na_2SO_4, and

concentrated under reduced pressure. Flash chromatography (SiO_2-NH, 40% AcOEt/*n*-hexane) afforded (+)-7-hydroxymitragynine (**3**, 22.0 mg, 71%). All spectral data of the obtained (+)-7-hydroxymitragynine (**3**) were identical to those of natural product.[6,8)]

(+)-7-Hydroxymitragynine (**3**): Pale yellow amorphous powder; $[\alpha]_D^{23}$ +62.5 (*c* 0.50, $CHCl_3$) [lit. $[\alpha]_D^{23}$ +47.9 (*c* 0.55, $CHCl_3$)]; IR (ATR) ν_{max} cm^{-1} 3420, 2947, 2872, 2835, 2811, 2745, 1698, 1644, 1597, 1486, 1460, 1435, 1376, 1266, 1237, 1187, 1143, 1104, 1074, 1018, 989, 959, 935, 920, 870, 841; HRMS (ESI) $[M+H]^+$ Calcd. for $[C_{23}H_{31}N_2O_5]^+$: 415.2233, Found: 415.2256; CD (0.3 mM, MeOH, 23 °C) λ nm ($\Delta\varepsilon$): 344 (0), 305 (0.53), 282 (0), 258 (0.82), 243 (0), 229 (−1.13), 214 (0); UV (MeOH) λ_{max} 307, 245, 220 nm; ^{1}H- and ^{13}C-NMR see supplementary materials.

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Conflict of Interest The authors declare no conflict of interest.

Supplementary Materials This article contains supplementary materials.

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